



R -A G (RAG):
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H

CONTEN

1. RAG O

2. RAG P

3. K E

4. RAG I P

5. P



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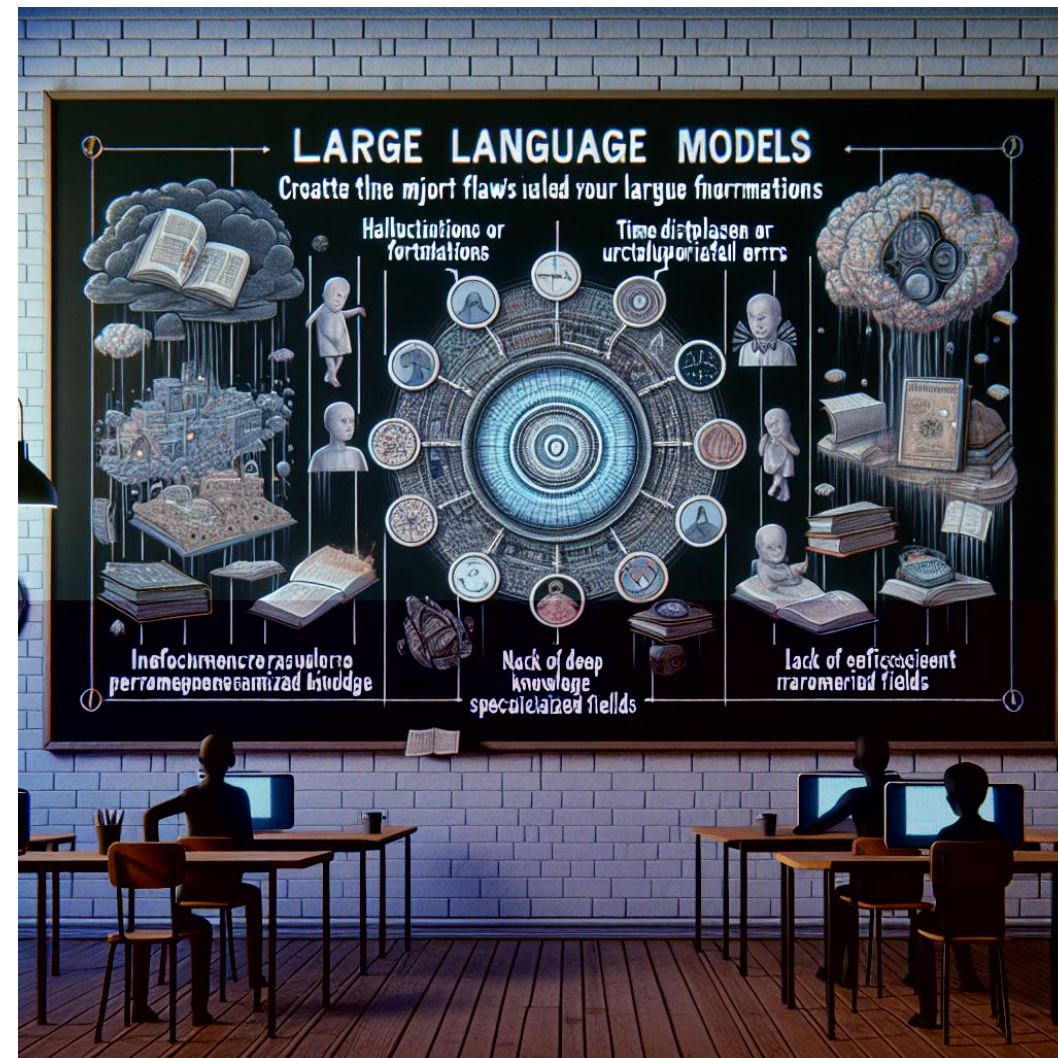
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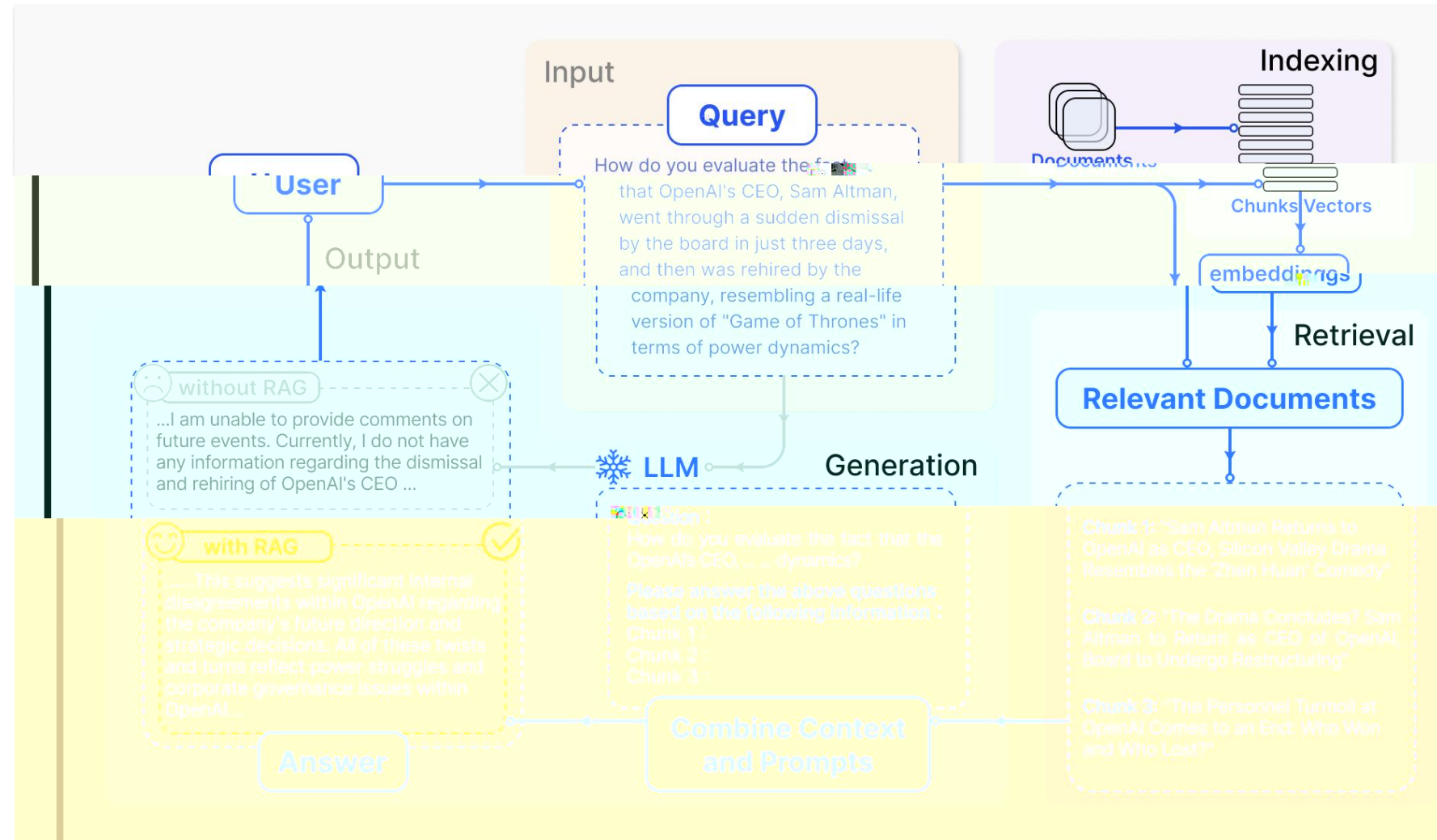
DALL E-3

➡ R -A G (RAG)

LLM

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RAG



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External Knowledge
Required

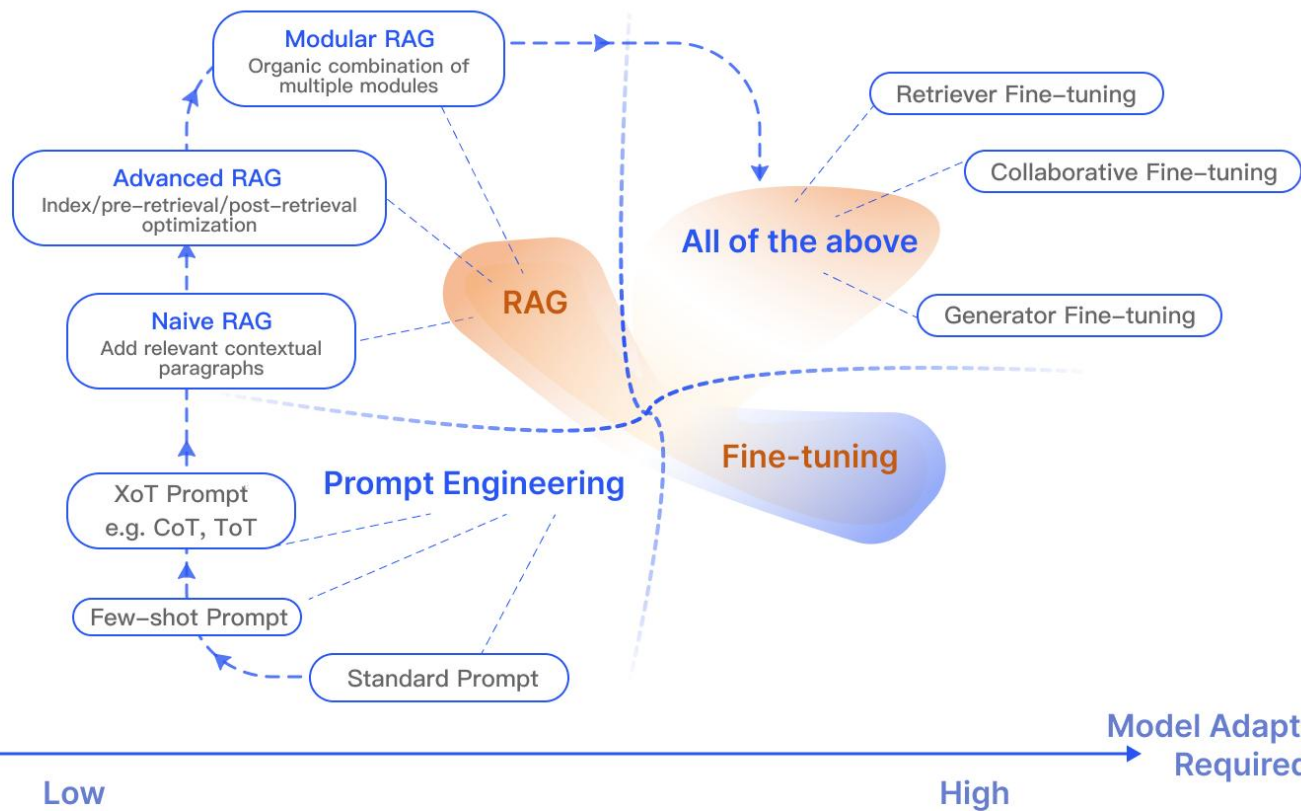
High

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High

Model Adaptation
Required



A

RAG

➡ RAG F -

Feature Comparison		RAG	Fine-Tuning
Knowledge Updates		Directly updating the retrieval knowledge base ensures that the information remains current without the need for frequent retraining, making it well-suited for dynamic data environments.	Stores static data, requiring retraining for knowledge and data updates.
External Knowledge		Proficient in leveraging external resources, particularly suitable for accessing documents or other structured/unstructured databases.	Can be utilized to align the externally acquired knowledge from pretraining with large language models, but may be less practical for frequently changing data sources.
Data Processing		Involves minimal data processing and handling.	Depends on the creation of high-quality datasets, and limited datasets may not result in significant performance improvements.
Model Customization		Focuses on information retrieval and integrating external knowledge but may not fully customize model behavior or writing style.	Allows adjustments of LLM behavior, writing style, or specific domain knowledge based on specific tones or terms.
Interpretability		Responses can be traced back to specific data sources, providing higher interpretability and traceability.	Similar to a black box, it is not always clear why the model reacts a certain way, resulting in relatively lower interpretability.
Computational Resources		Depends on computational resources to support retrieval strategies and technologies related to databases. Additionally, it requires the maintenance of external data source integration and updates.	The preparation and curation of high-quality training datasets, defining fine-tuning objectives, and providing corresponding computational resources are necessary.
Latency Requirements		Involves data retrieval, which may lead to higher latency.	LLM after fine-tuning can respond without retrieval, resulting in lower latency.
Reducing Hallucinations		Inherently less prone to hallucinations as each answer is grounded in retrieved evidence.	Can help reduce hallucinations by training the model based on specific domain data but may still exhibit hallucinations when faced with unfamiliar input.
Ethical and Privacy Issues		Ethical and privacy concerns arise from the storage and retrieval of text from external databases.	Ethical and privacy concerns may arise due to sensitive content in the training data.

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RAG :

RE RO (B 2021)
REALM (G , 2020)
A LA (, 2023)

RAG (L , 2020)
A LA (, 2022)
E . G (A , 2022)

B B 3 (.2022)
I -
(K ., 2022)

FLARE (J , 2023)

NN-M (K .,
2020) RIME-M (.,
2022)

D P (.,
2023
N P .,
2022)

NN-P (., 2022)
NPM (M ., 2023)

NN-P (., .,
2022)NPM (M ., 2023)

R (, 2022)

PAR 02
RAG P



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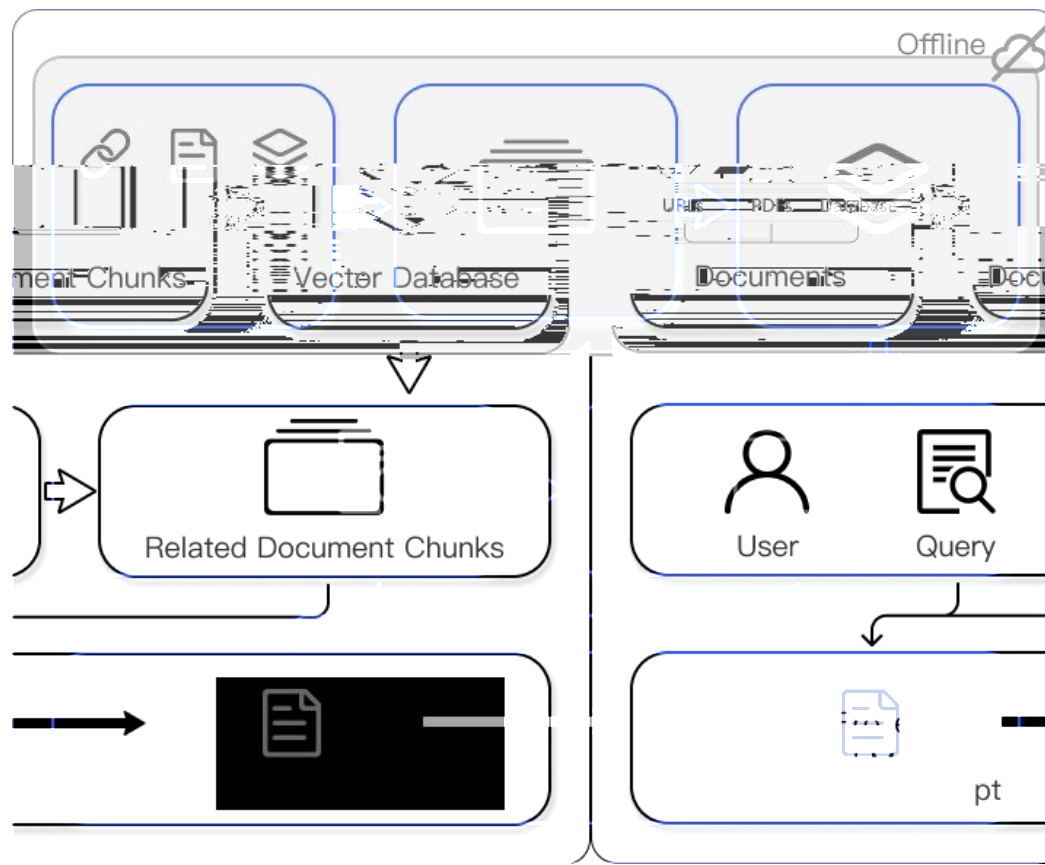
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3 G

LLM



A RAG



M RAG

▶ A RAG

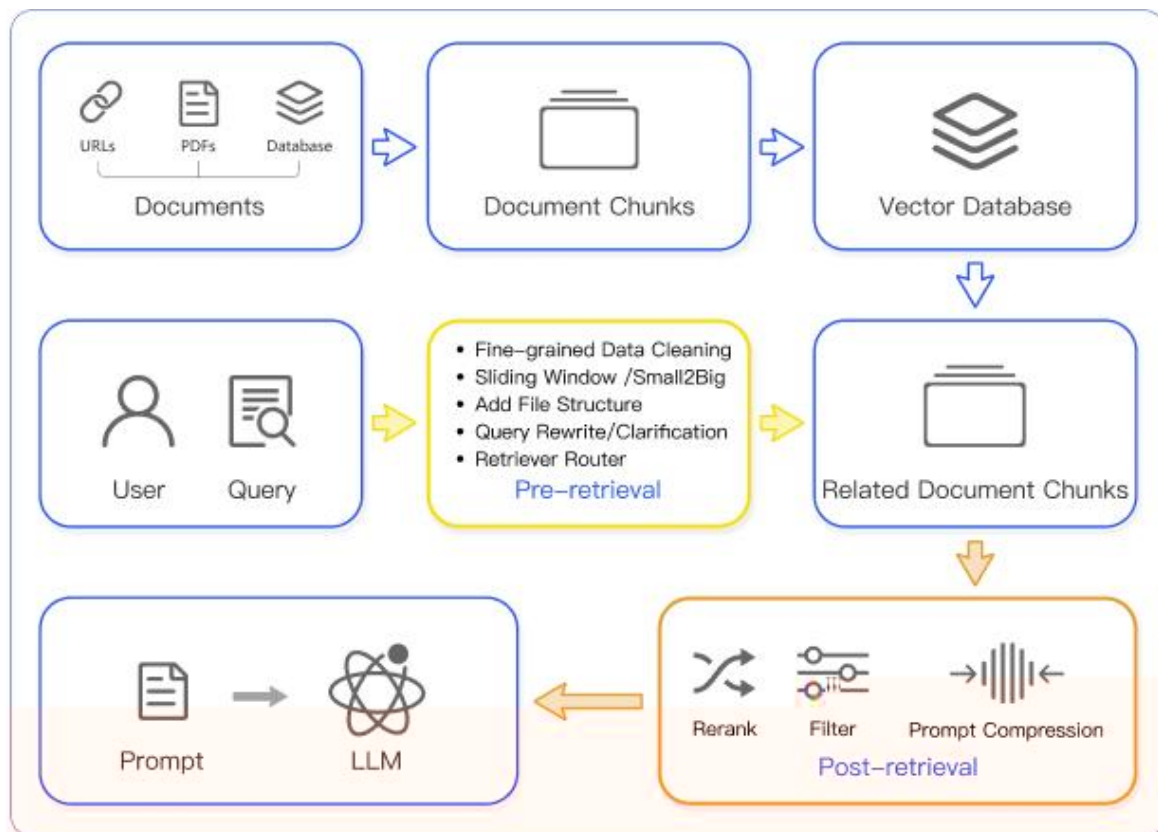
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P -R P → G

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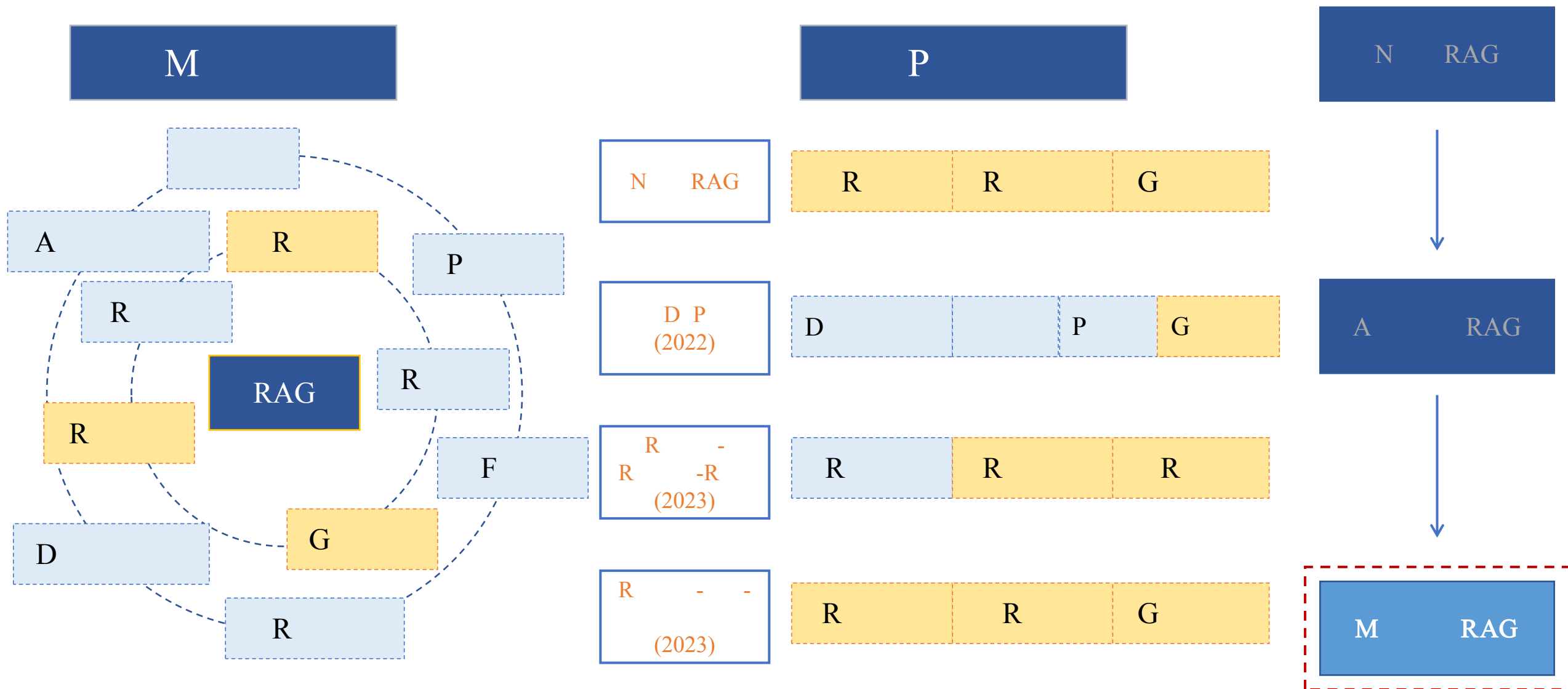


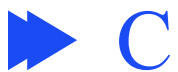
N RAG



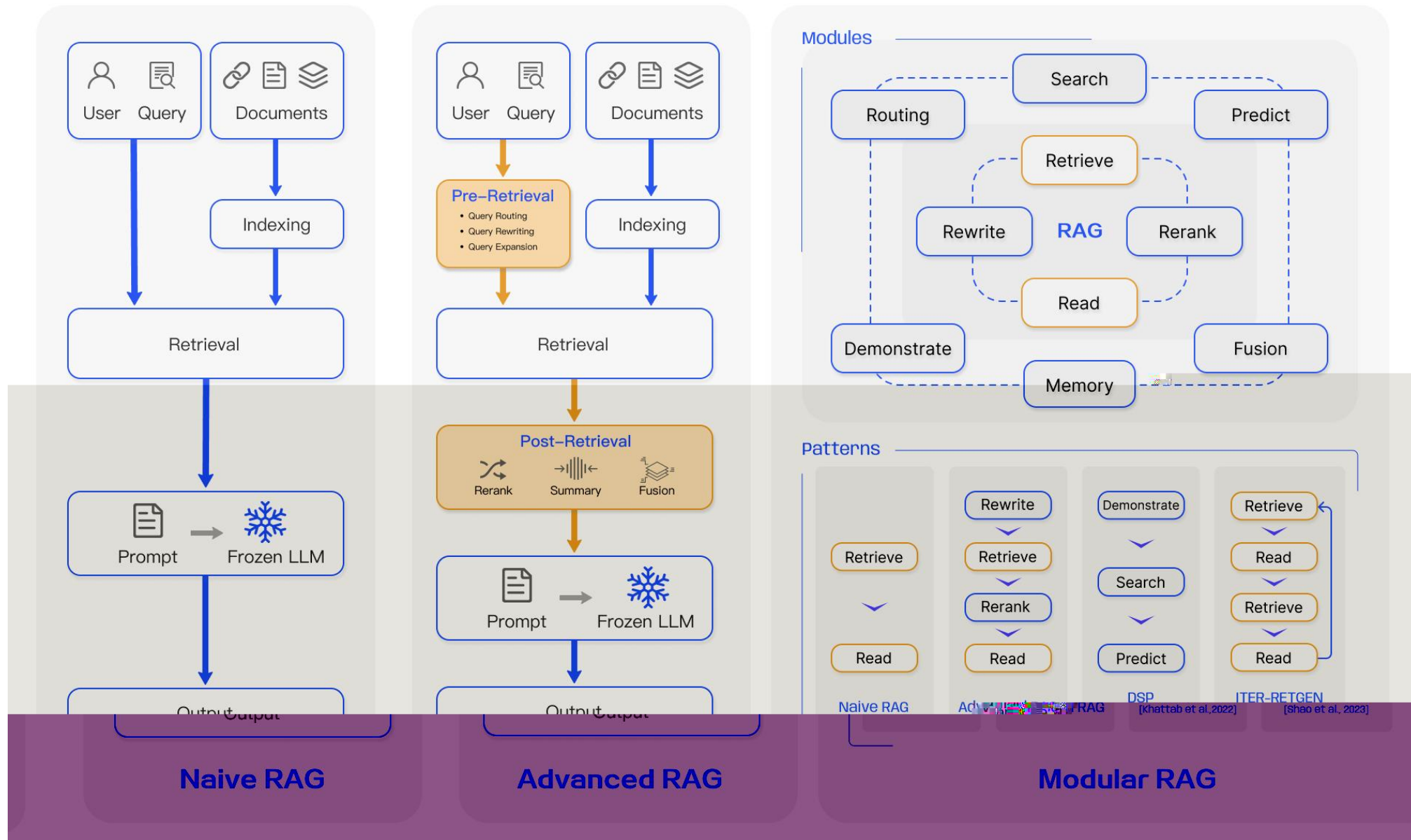
M RAG

► M RAG





RAG P





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M /I L
O /P L

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I

R



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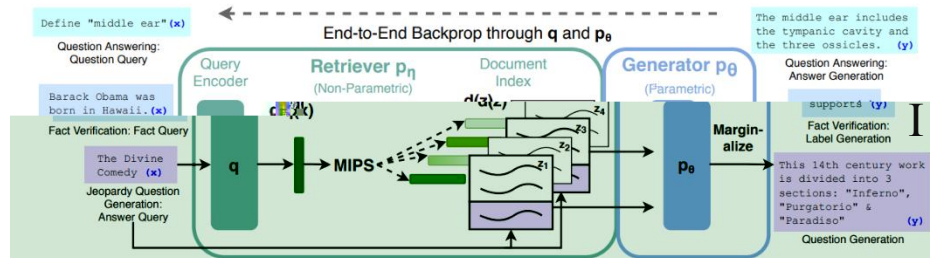
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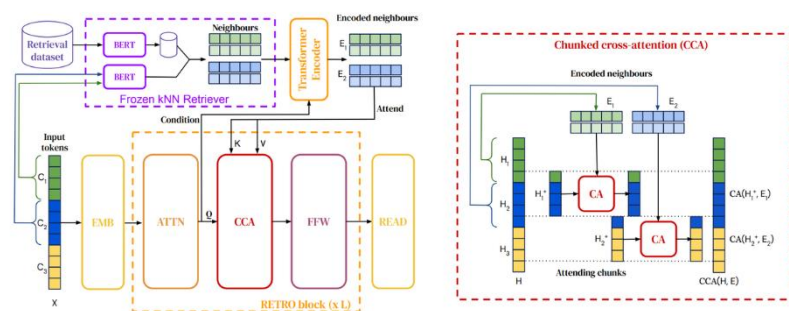
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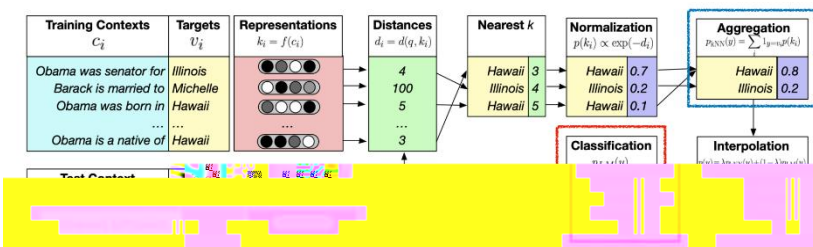
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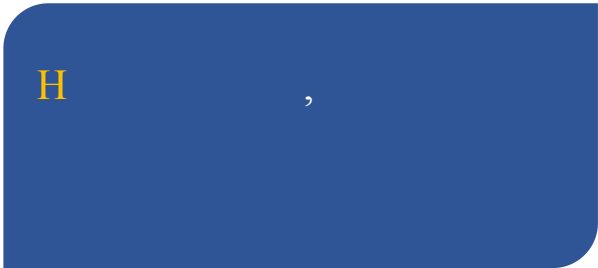
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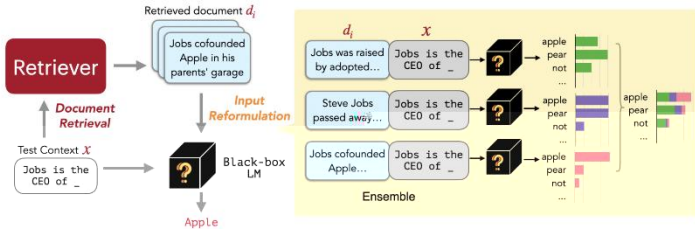
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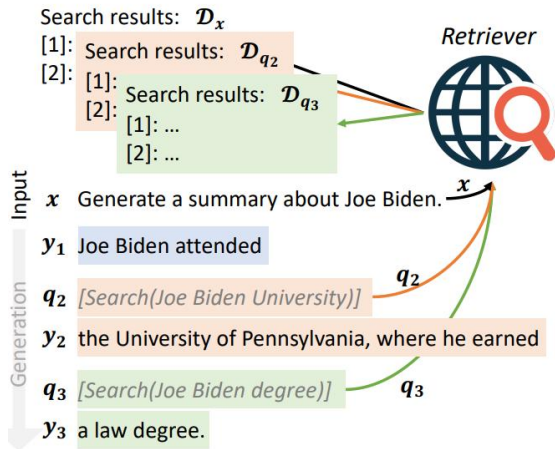
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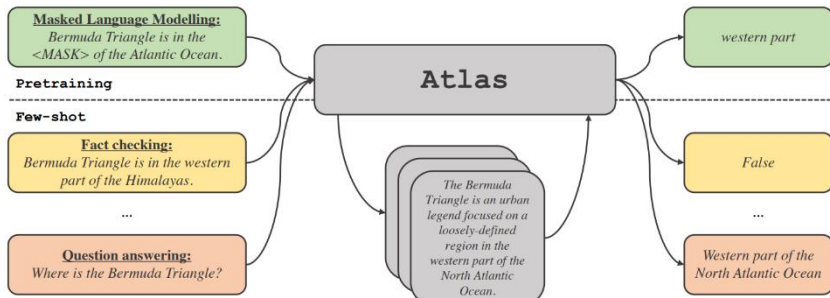
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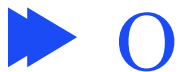
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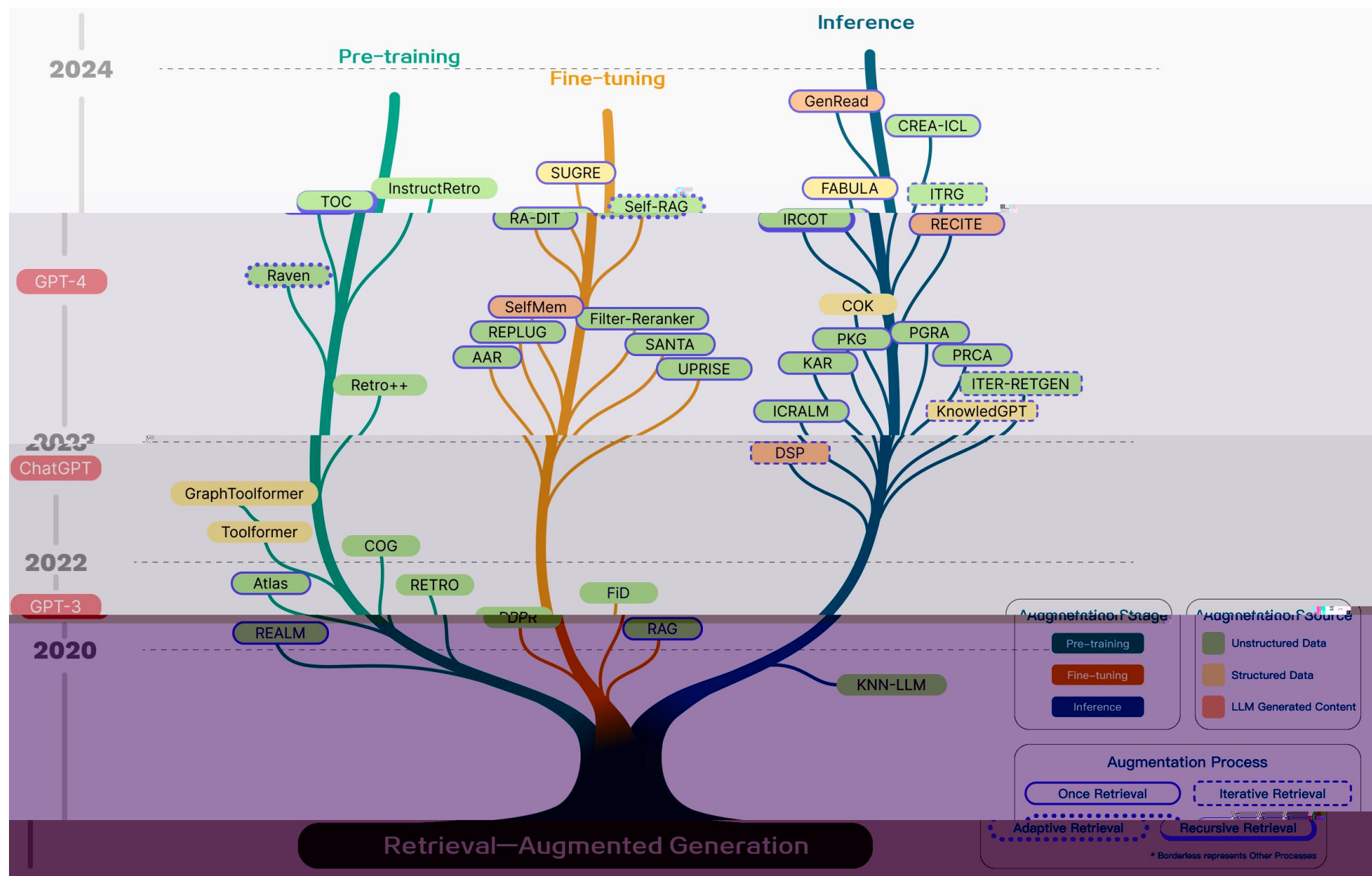
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RAG D



PAR 03
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B R G D

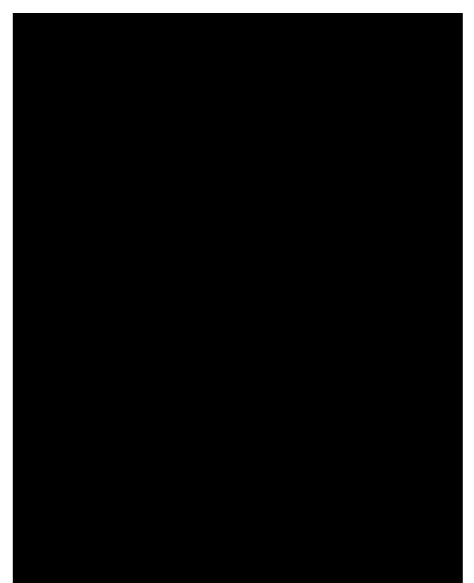
[Embed Sentence → Link to Expanded Window](#)

Continuous observation of the Atlantic meridional
turning circulation (AMOC) has improved the
understanding of its variability (Frajka-Williams et al.,
2019). However, there is low confidence in the quantification
of changes in the 20th century because of low
observational records since the
quantitative reconstructed and simulated
observational records since the
too short to determine the relative
internal variability, natural
anthropogenic forcing to AMOC change
the 21st century, AMOC will
scenarios but will not
before 2100. 3.2.2.4 Sea Ice

What the LLM Sees

What the LLM Sees

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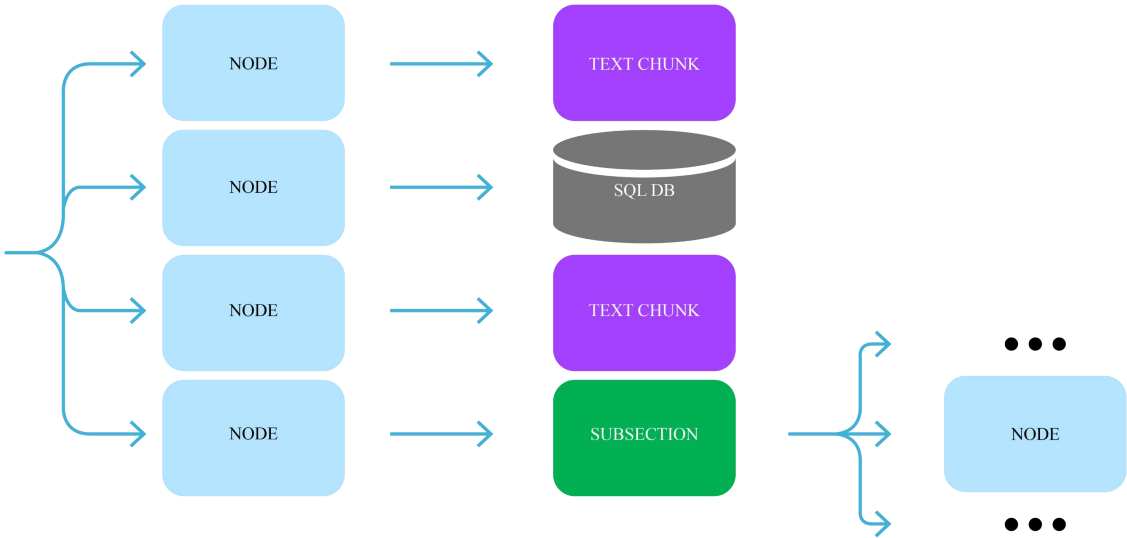
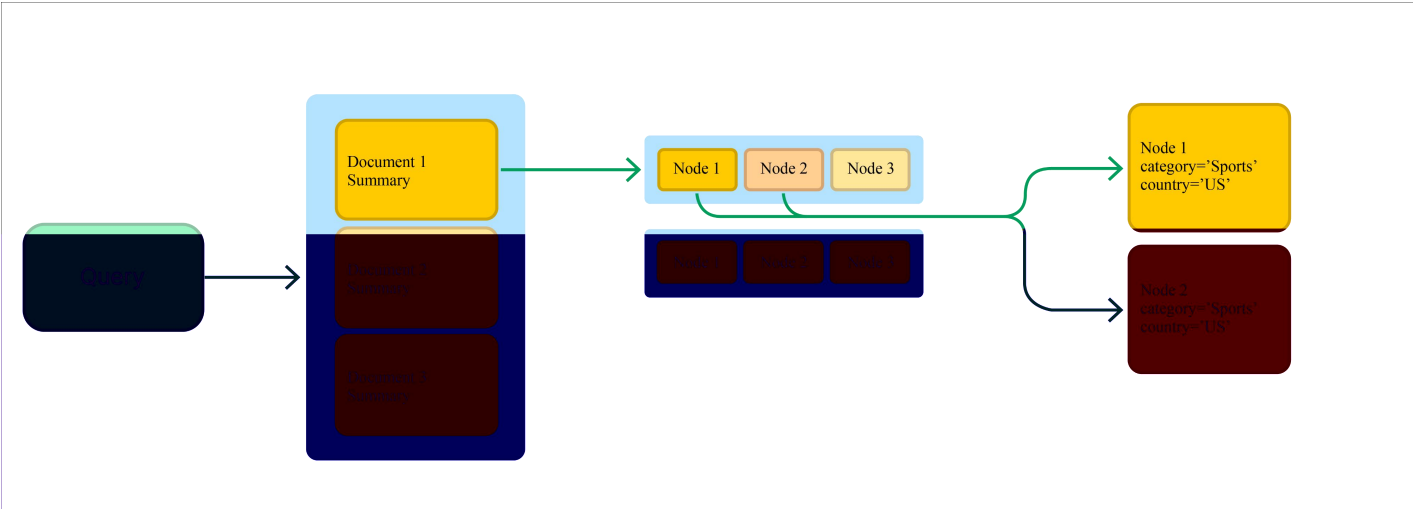




B RAG

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H O R C





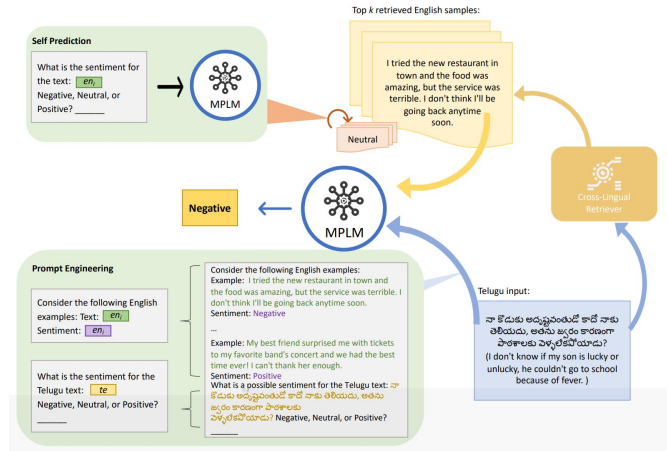
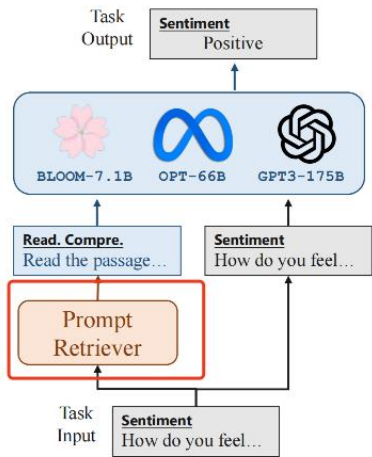
B RAG R O

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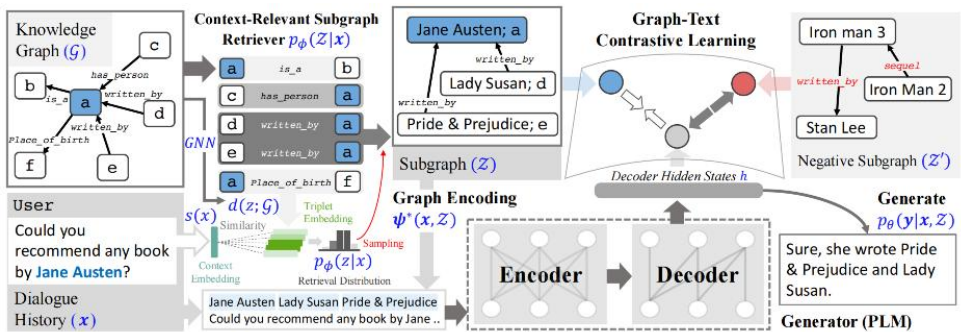
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P P R I E C ., 2023 C - C R E A - I C L L ., 2023

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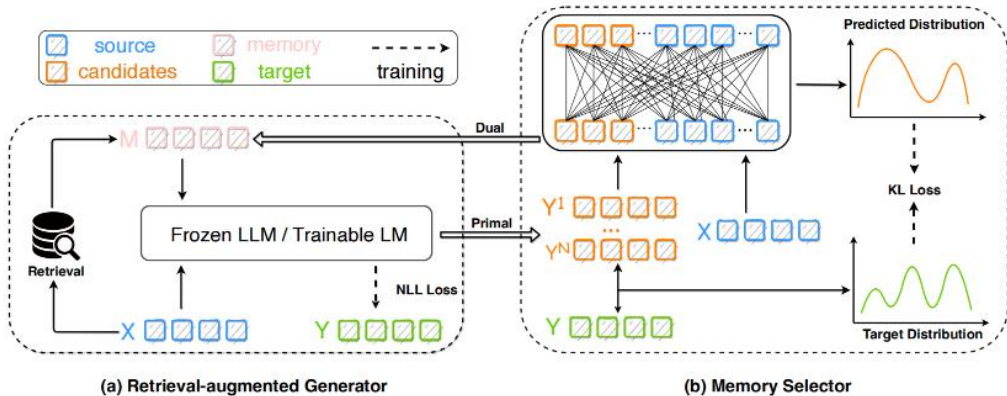
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LLM M

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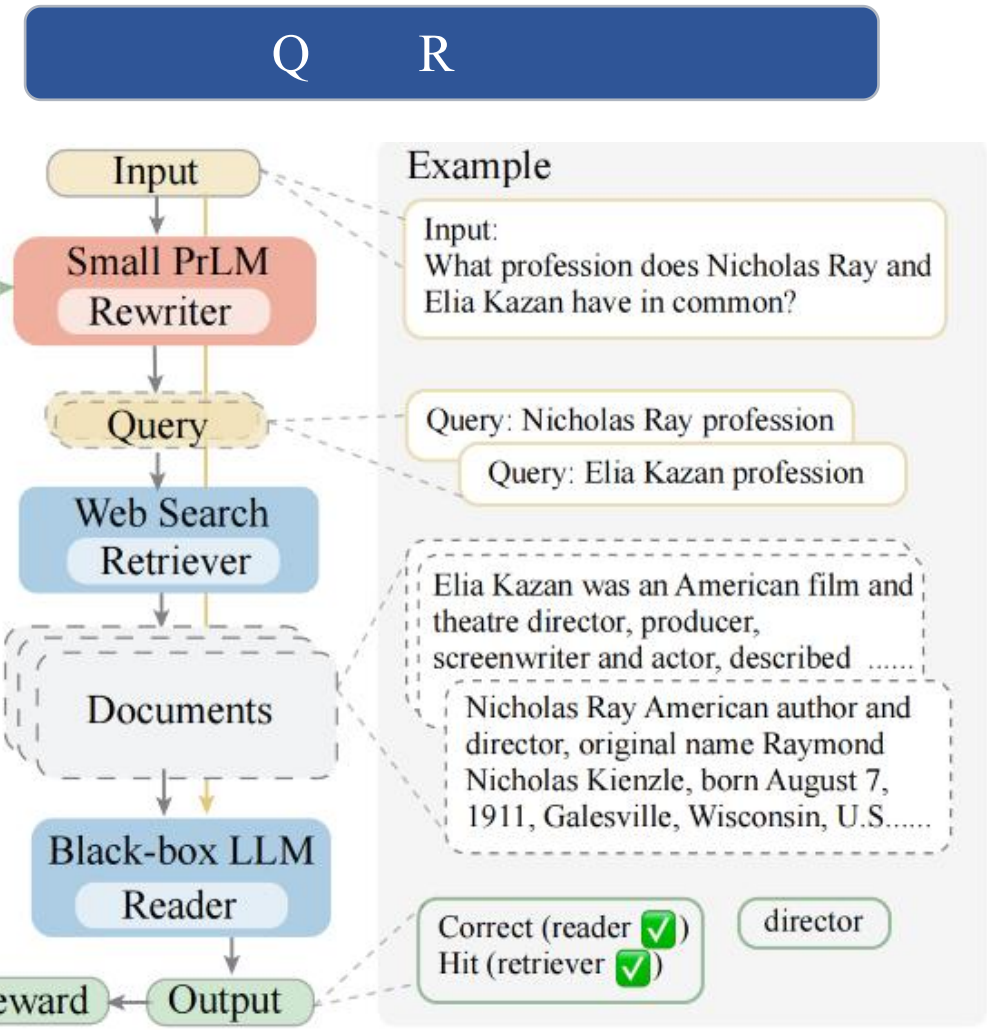


M C ., 2023



B RAG

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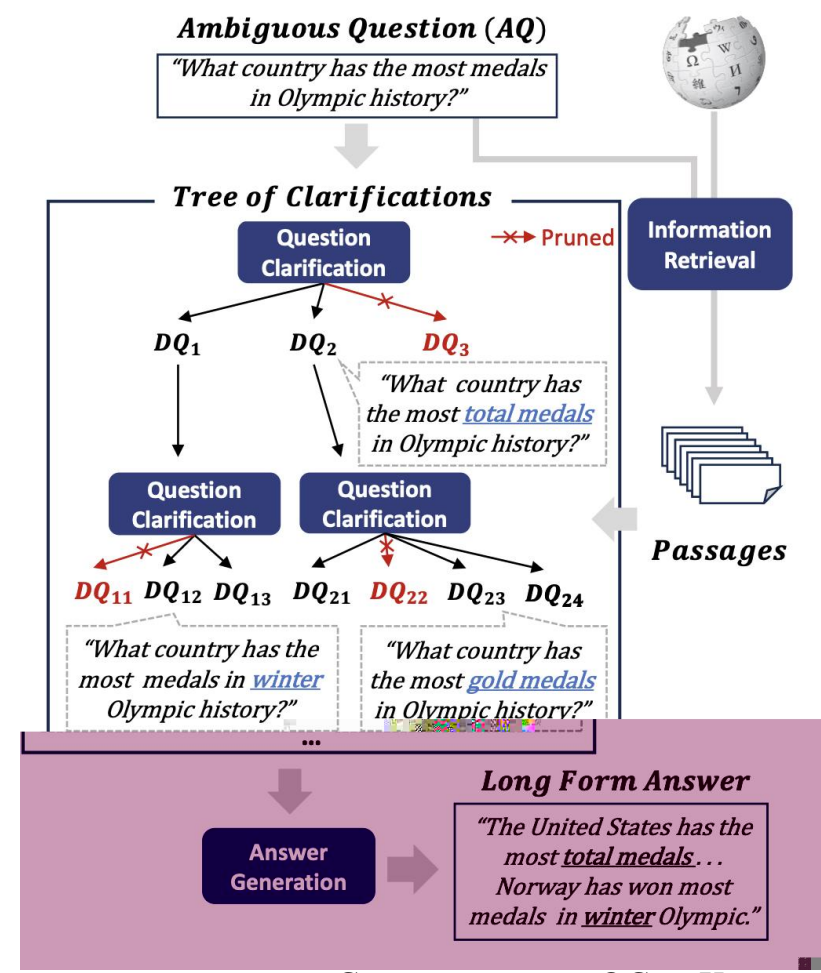
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C OC K ., 2023



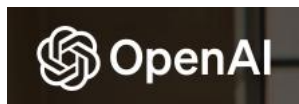
B RAG E

E M P



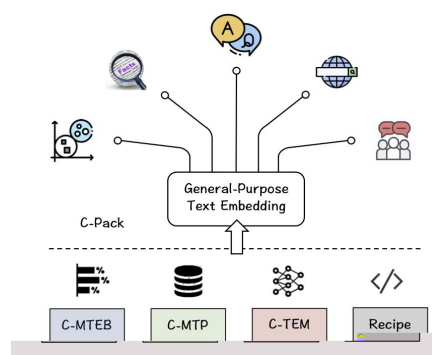
M M3E

VOYAGE AI



llmrails

BAAI
智源研究院



BAAI-G

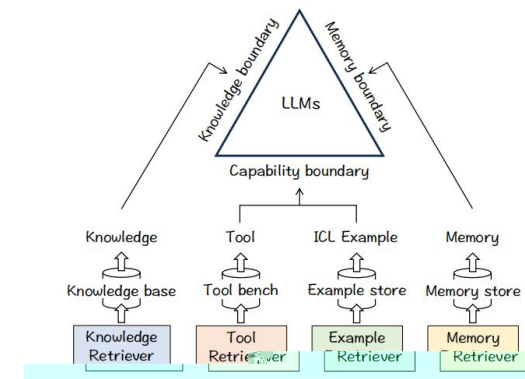
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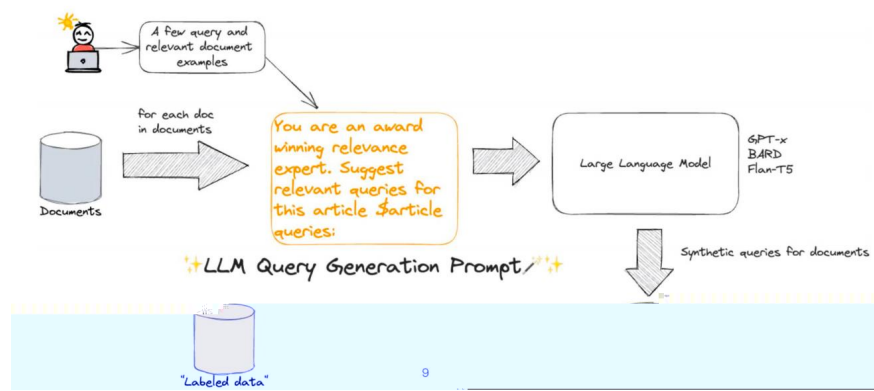
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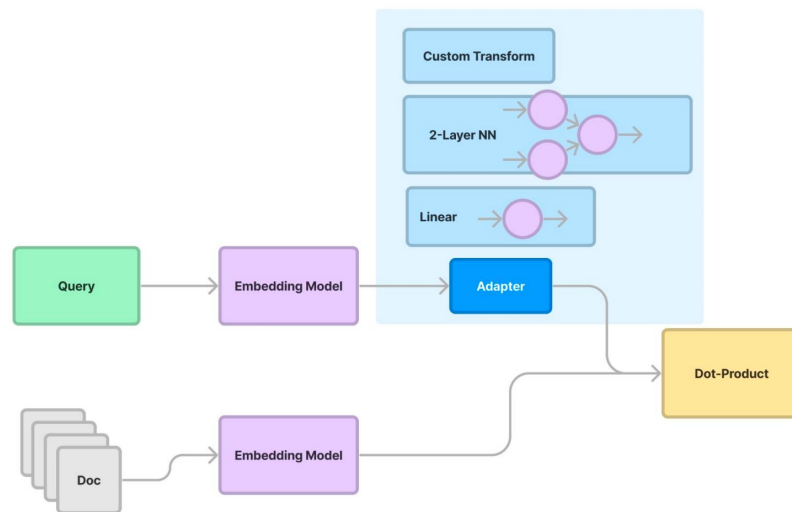
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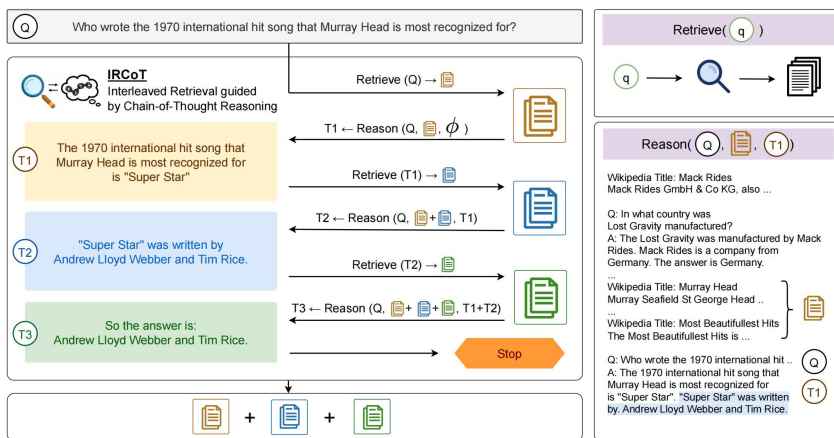
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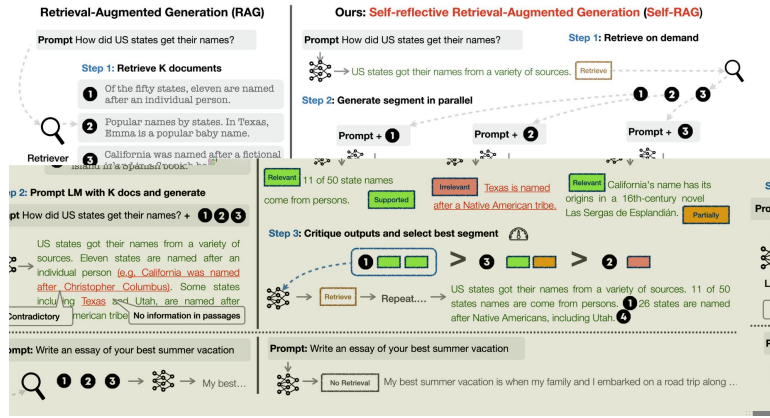
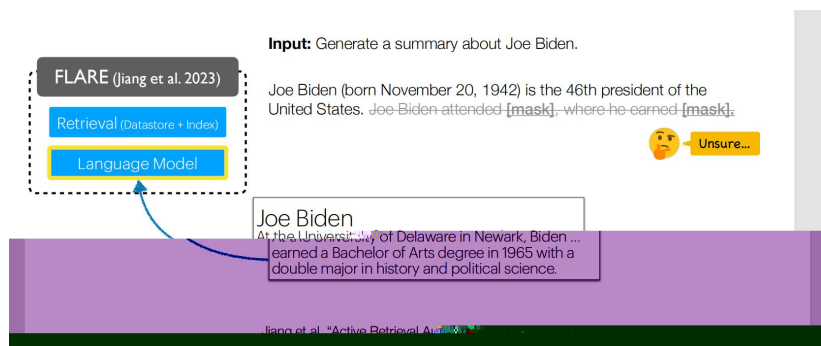


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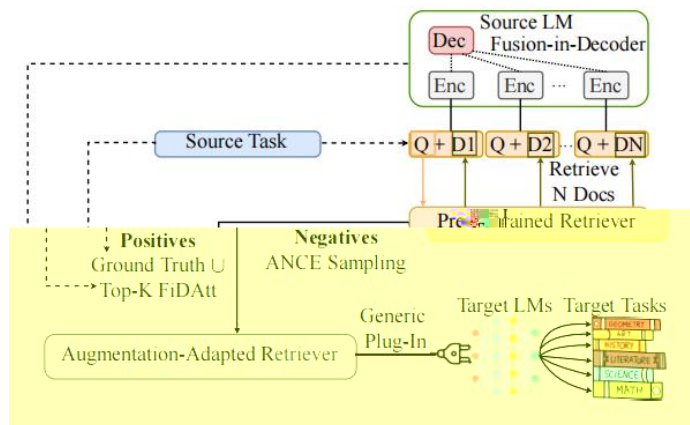
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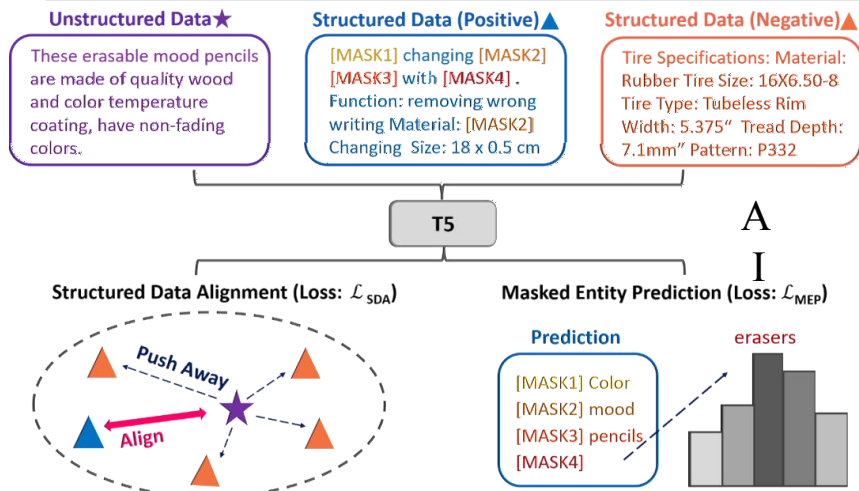
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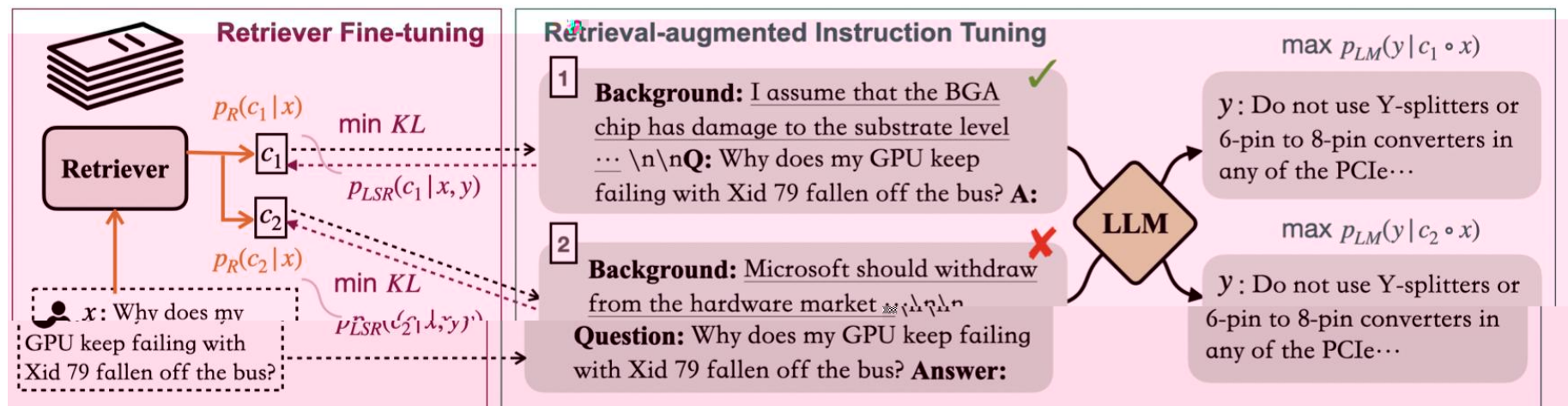
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Structured Data Alignment (Loss: $\mathcal{L}_{SDA}$)Masked Entity Prediction (Loss: $\mathcal{L}_{MEP}$)

Prediction

AN A L

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M :MRP, H R ,NDCG

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R D E
M :C R

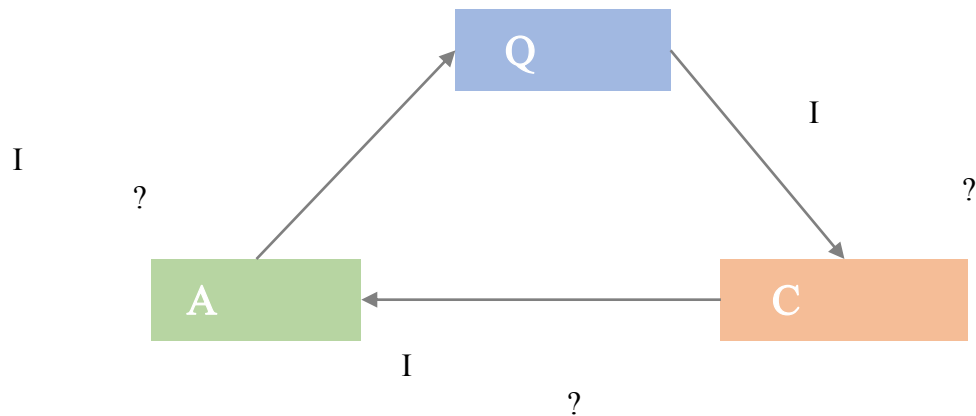
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- A F
- A R
- C R

PAR 04

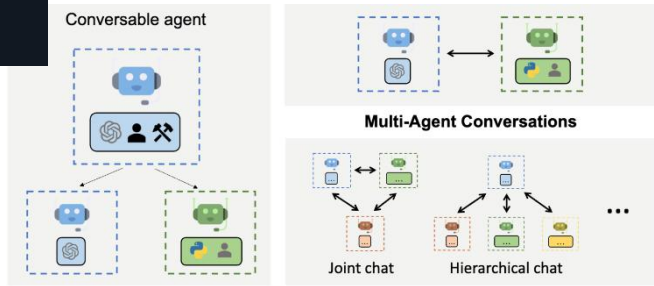
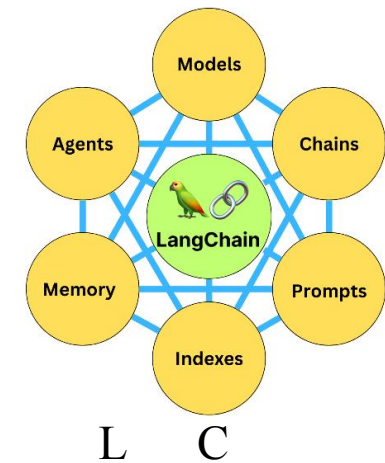
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F	AI	E	,	D	
A	G	A	-	L	

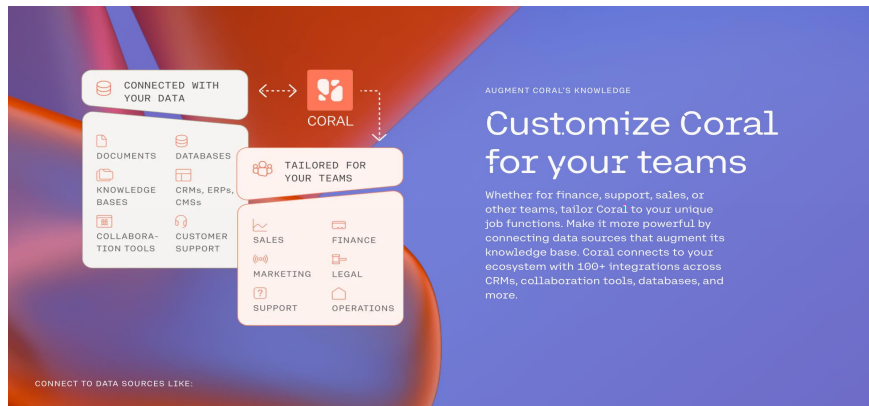


A G

▶ RAG I A P



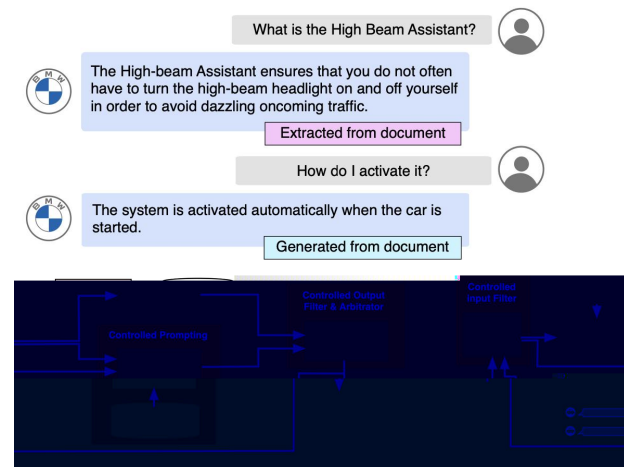
N E - C B I



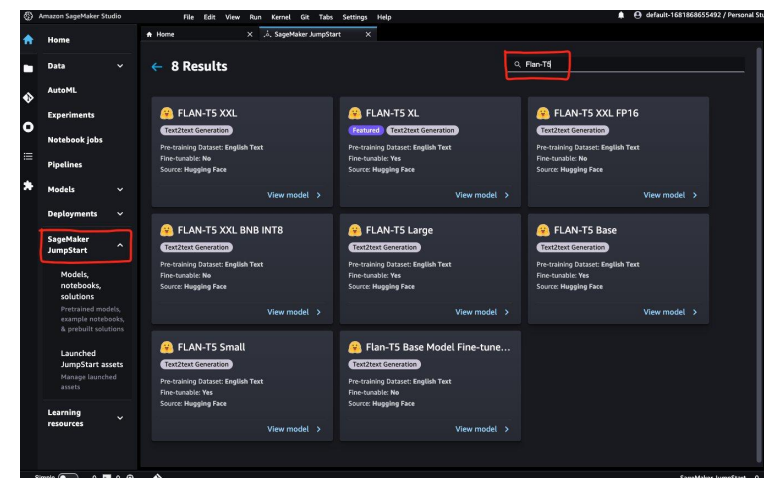
C - C



AI
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BM - C E



A - K

PAR 06

O



F

RAG

» RAG Ecosystem

Downstream Tasks

Dialogue
Question answering
Summarization
Fact verification

Technology Stacks

Langchain
LlamaIndex
FlowiseAI
AutoGen

» The RAG Paradigm

Naive RAG → Advanced RAG → Modular RAG

» Techniques for Better RAG

Chunk Optimization
Iterative Retrieval
Retriever Fine-tuning
Query Rewriting
Recursive Retrieval
Generator Fine-tuning
Rerank
Adaptive Retrieval
Dual Fine-tuning

» Key Issues of RAG

What to retrieve
When to retrieve
How to use Retrieval

» RAG Prospect

Challenges

Context Length
Robustness
Hybrid
Role of LLMs
Scaling-laws for RAG
Production-ready RAG

Modality Extension

Image
Audio
Video
Code

Ecosystem

Customization
Simplification
Specialization

» Evaluation of RAG

Evaluation Target

Retrieval Quality
Generation Quality

Evaluation Aspects

Answer Relevance
Context Relevance
Answer Faithfulness
Noise Robustness
Negation Rejection
Information Integration
Counterfactual Robustness

Evaluation Framework

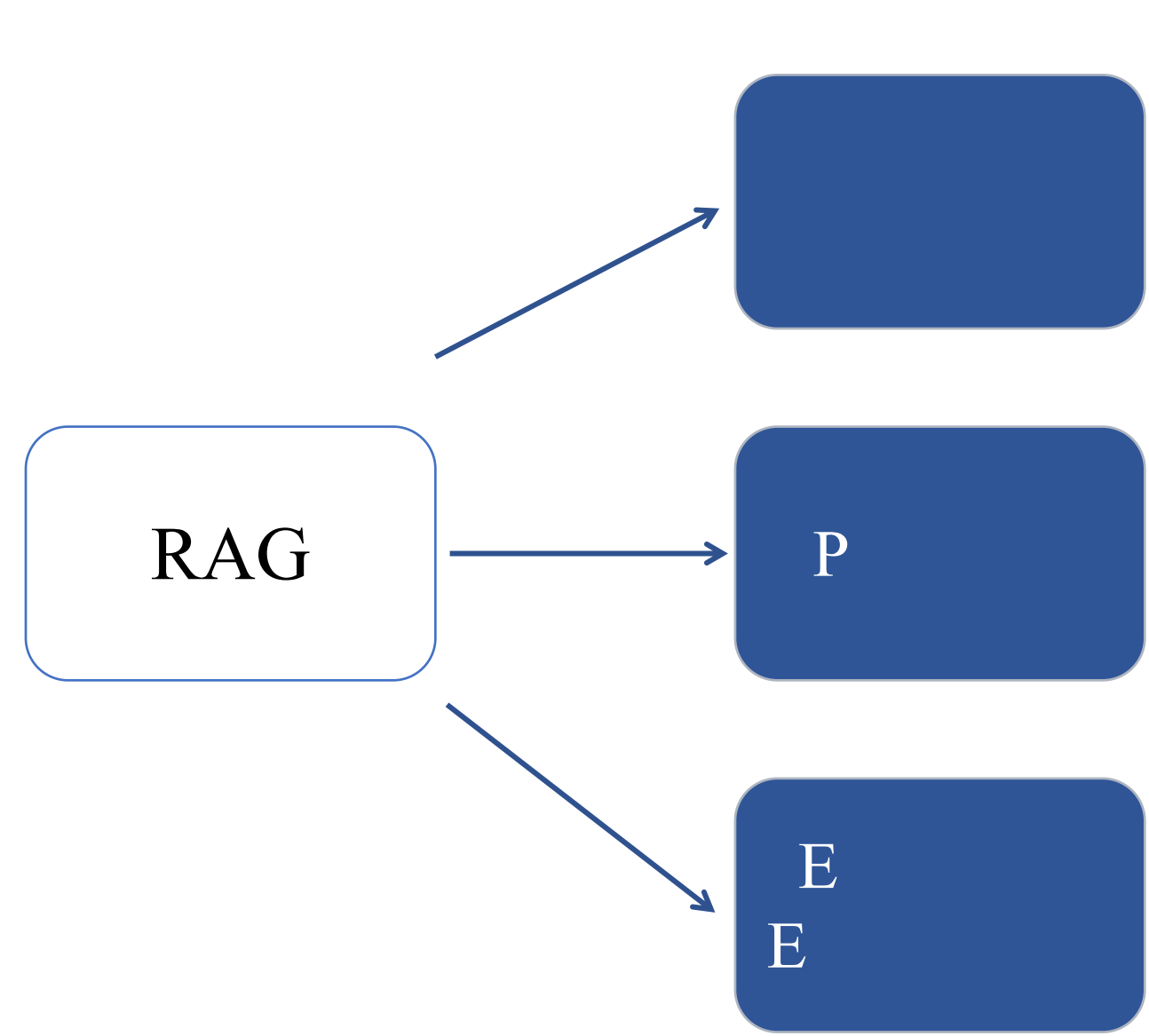
Benchmarks

Tools

RGB
RECALL
TruLens
RAGAS
ARES



RAG



L RAG M
H I E R
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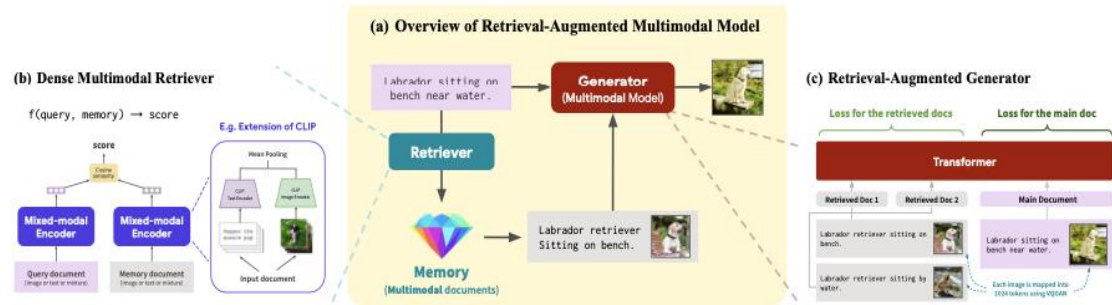
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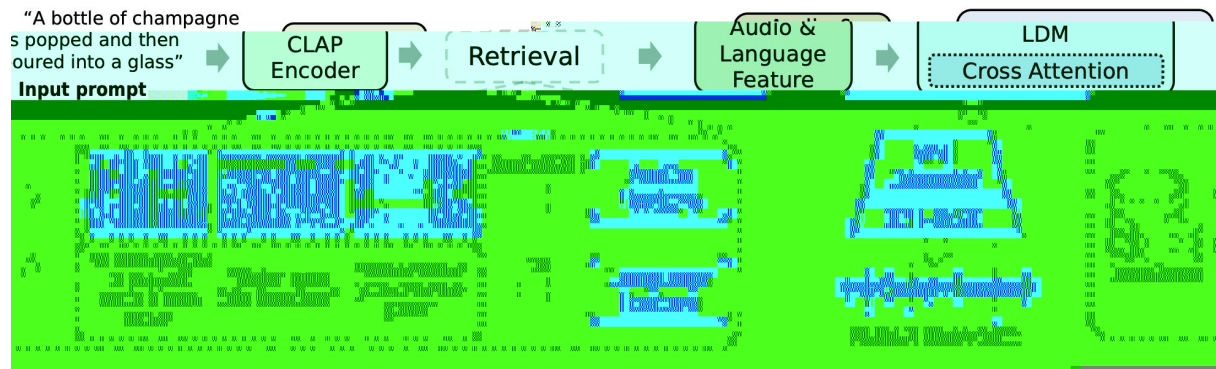
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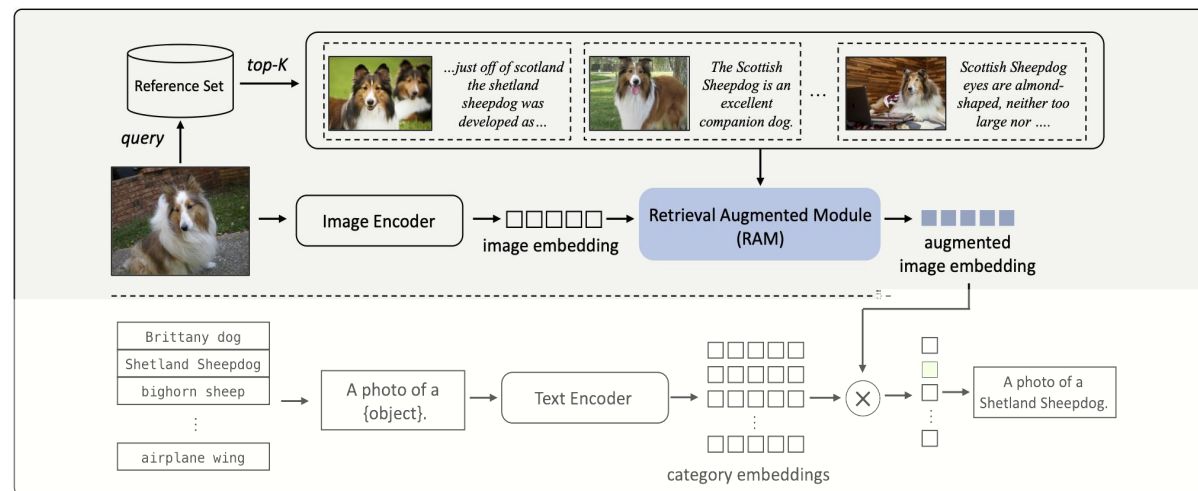
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R -A

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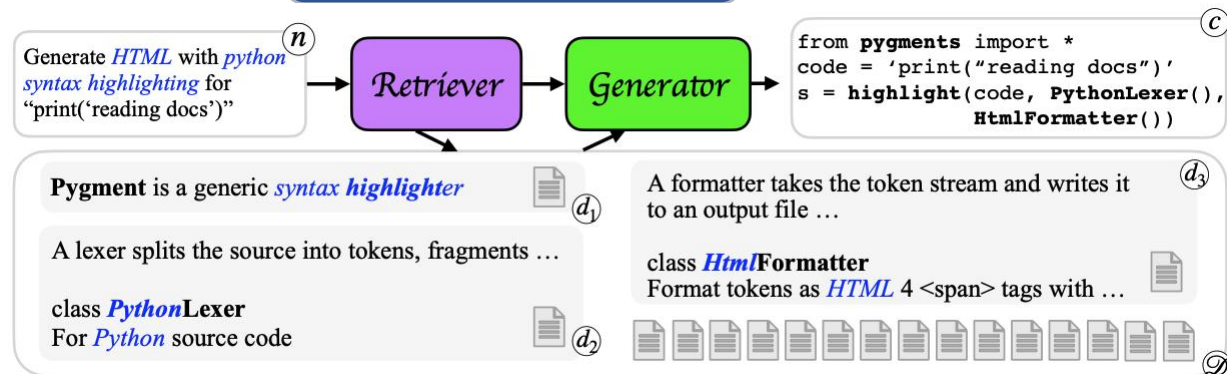
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RA-CLIP

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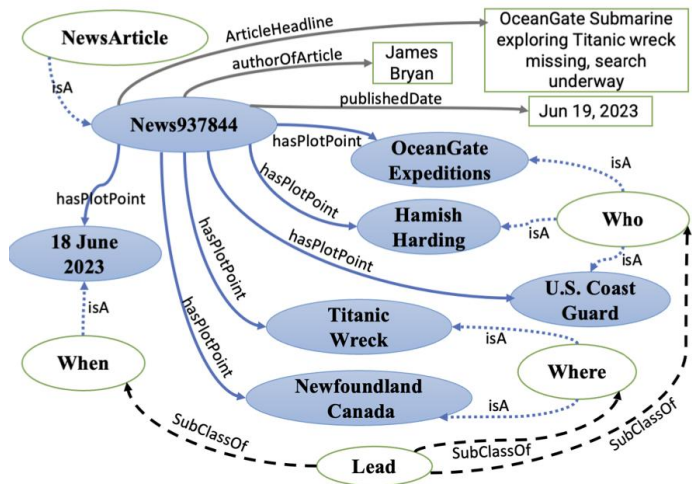
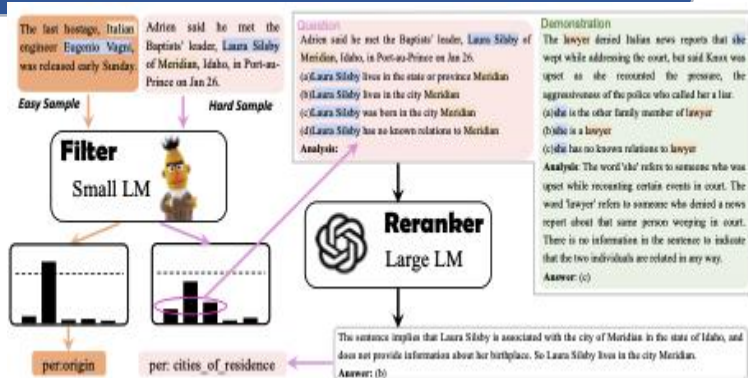
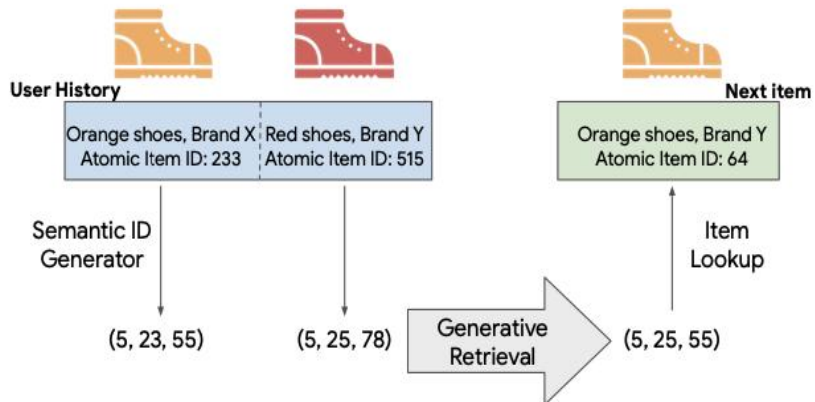
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Verba

The Golden RAGtriever



haystack
by deepset

P
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RAG

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1. A , . .N - L M A - R .
2. L , P. .R -A G K -I NLP .
3. G , K., L , K., , ., P , P. & C , M.- .REALM: R -A L M P - . P
:// . / /2002.08909 (2020).
4. D , . .P :F - D R F 8 E .P :// . / /2209.11755 (2022).
5. I , G. .A :F - L R A L M .P :// . / /2208.03299 (2022).
6. G , L., M , ., L , J. & C , J. P - D R R L .P :// . / /2212.10496 (2022).
7. M , N., , N., M , L. & R , N. M EB: M E B . P 17 C E
C A C L 2014 2037 (A C L , 2023).
8. R , . .R - - :D - E A E H R A . P 61 A
M A C L (1: L P) 293 306 (A C L , 2023).
9. , J. .R A KD: R -A K D F P - L M . P 61 A M
A C L (2: P) 1128 1136 (A C L , 2023). 10. K , O. .
D - -P :C - NLP. P :// . / /2212.14024 (2023).
11. C , . .L :R - G M .P :// . / /2305.02437 (2023).
12. L , . .A L L M P K G .P :// . / /2305.04757 (2023).
13. , . .REPL G: R -A B -B L M .P :// . / /2301.12652 (2023).
14. , ., , C., , . & L , .A -A R I G L M G P -I .P
:// . / /2305.17331 (2023).
15. K , M., K , J. M., B , J. & H , . J. K G -A L M K -G D G . P
:// . / /2305.18846 (2023).
16. , H., B , N., K , . & , A. I R C - - R K -I M -
Q .P :// . / /2212.10509 (2023).
17. , L., , N. & , F. L R I -C E L L M .P :// . / /2307.07164 (2023).
18. L , . . G E M - C L .P :// . / /2308.03281 (2023).
19. N , . . R :A P L R -C G AI .P :// . / /2308.03983 (2023).
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21. , . .L L M I R :A .P :// . / /2308.07107 (2023).
22. , . .K GP :E L L M R A K B .P :// . / /2308.11761 (2023).
23. C ,J., L ,H., H , . & ,L. B L L M R -A G .P :// . / /2309.01431
24. E , .,J ,J., E -A ,L. & , . RAGA :A E R A G .P :// . / /2309.15217 (2023).
25. ,O., , .,R ,O. & B ,J. M R -A L M R I C .P :// . / /2310.01558 (2023).
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27. ,H. . B :E R A L L M .P :// . / /2310.06117 (2023).
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